Urban Flooding and Adaptation

Allan Hsiao
Stanford

Dev Patel
Harvard

March 6, 2026

This paper

- Spatial sorting by income
  - Compensating differentials for environmental risk
- Adaptation policy faces a trade-off
  - Protecting risky areas reduces inequality in exposure
  - But also reduces housing affordability
- Today: theory and data from Indian cities

Residential choice

$$v_{ij} = -\alpha_j p_i - \beta_j e_i$$

- Locations $i \in \{$High risk, Low risk$\}$
- Groups $j \in \{$Poor, Rich$\}$

- Location utility v_{ij}
  - Housing prices p_i
  - Flood exposure $e_i = n_i - g_i$
  - Natural n_i, where $n_H > n_L$
  - Infrastructure g_i

Equilibrium prices

$$D_i(\hat{p}) = \sum_j \pi_{ij}(\hat{p}), \quad S_i = 1$$

- Demand from logit choice probabilities π_{ij}
- Supply from fixed housing stock S_i

Compensating differentials

$$\underbrace{\Delta \hat{p}}_{\hat{p}_H - \hat{p}_L} = -\bar{\gamma} \Delta e, \quad \bar{\gamma} = \frac{\beta_P + \beta_R}{\alpha_P + \alpha_R}$$

- Higher risk in H, lower price in H
- Market clears on aggregate $\bar{\gamma}$ (WTP to avoid flooding)

Sorting by income

$$c = \log\left(\frac{\pi_{HP}}{\pi_{LP}}\right), \quad \frac{\mathrm{d}\hat{c}}{\mathrm{d}\Delta e} = \alpha_P(\bar{\gamma} - \gamma_P)$$

- Sorting: concentration c of the poor in H, given exposure gap Δe
  - The rich crowd out the poor in pursuit of higher ground (Hsiao 2024)
  - Worsens inequality in exposure

- For $\gamma_j = \beta_j / \alpha_j$, sorting iff

$$\boxed{\bar{\gamma} > \gamma_P} \quad \text{and} \quad \bar{\gamma} < \gamma_R$$

Welfare by income

$$\frac{\mathrm{d}\hat{U}_j}{\mathrm{d}\Delta e} = (\bar{\gamma} - \gamma_j)\pi_{Hj}$$

- Under sorting, $\boxed{\bar{\gamma} > \gamma_P}$ and $\bar{\gamma} < \gamma_R$
  - The poor gain from exposure differences; the rich lose
  - Reduces inequality in welfare

- Market clears on aggregate $\bar{\gamma}$ (WTP)
  - For the poor, over-compensation through housing prices

Adaptation policy faces a trade-off

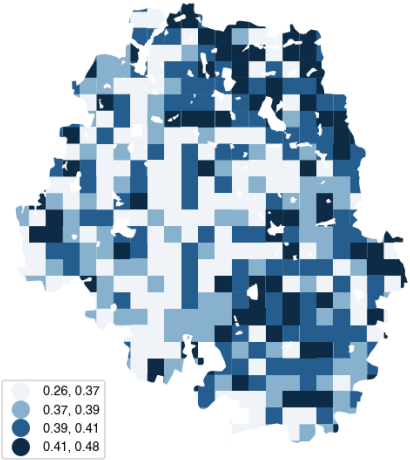
- In-situ and ex-situ adaptation interact
  - Policy: government infrastructure (in situ)
  - Response: migration and sorting (ex situ)
- Exposure-based allocation protects H ($n_H > n_L$)
  - Exposure gap $\downarrow$, aggregate losses $\downarrow$
  - But have gentrification and uniformly expensive housing
- Price-based allocation protects L ($\hat{p}_L > \hat{p}_H$)
  - Exposure gap $\uparrow$, aggregate losses $\uparrow$
  - But preserves pockets of affordable housing

New data on flooding in Indian cities

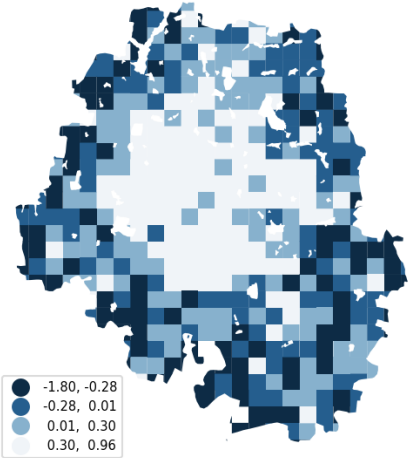
- Synthetic aperture radar to measure flooding
  - Impervious to weather, works at night
  - Define flooding as deviation from norm in surface water extent
- Can incorporate more granular satellite data (Patel 2024)
  - With customization for urban environments
- Today: data for Bangalore
  - Relative Wealth Index from Meta Data for Good
  - Stormwater drains from OpenCity India

Flooding and wealth

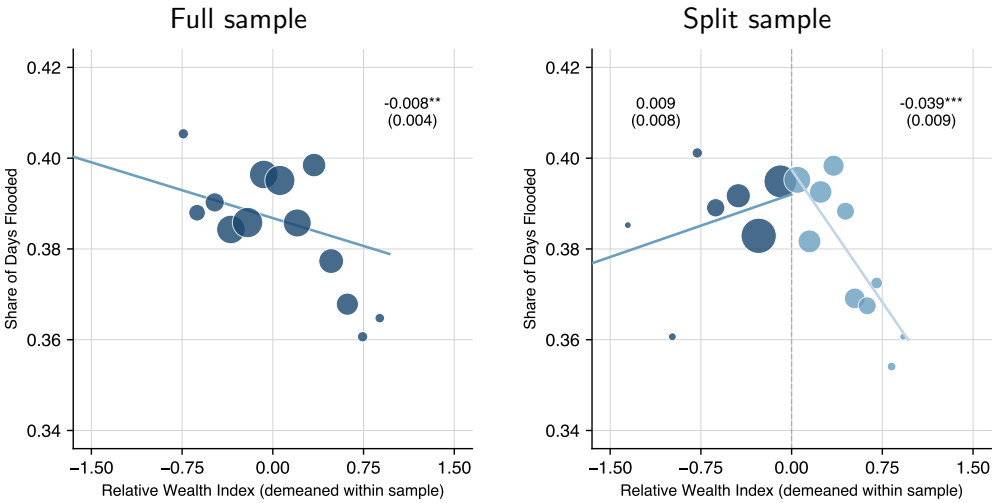
Share of days flooded



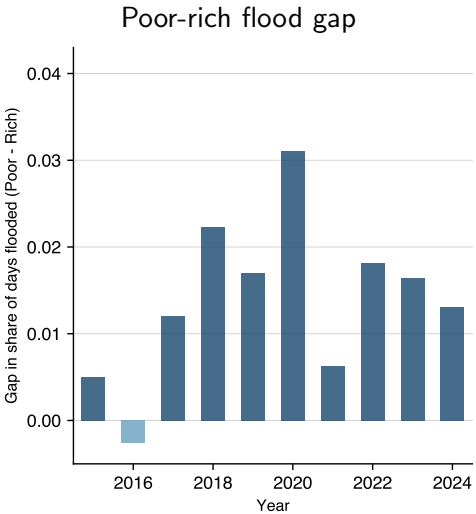
Relative wealth index

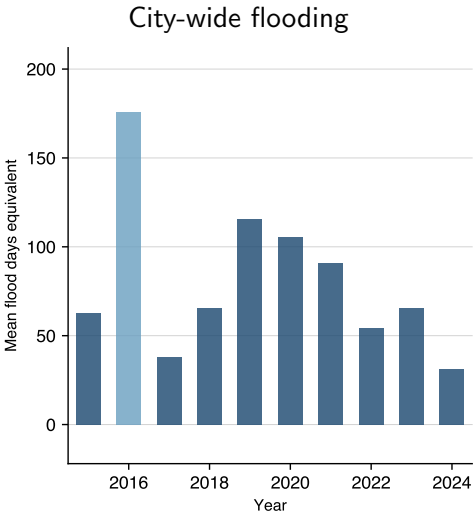


Flooding and wealth across space



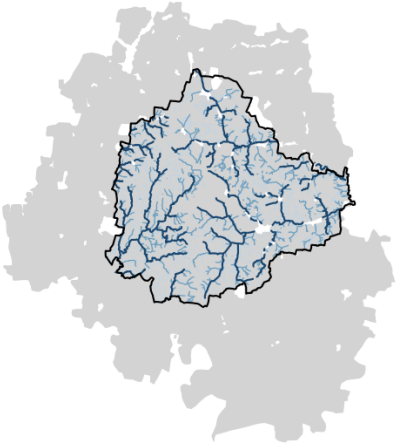
Flooding and wealth over time



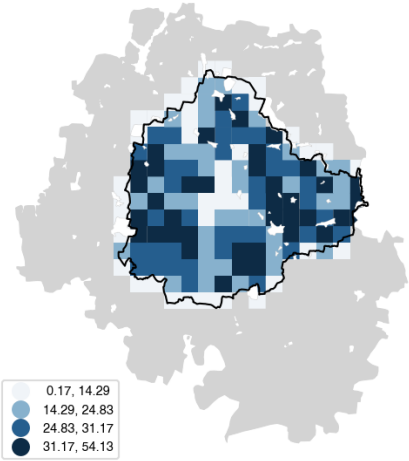


Infrastructure

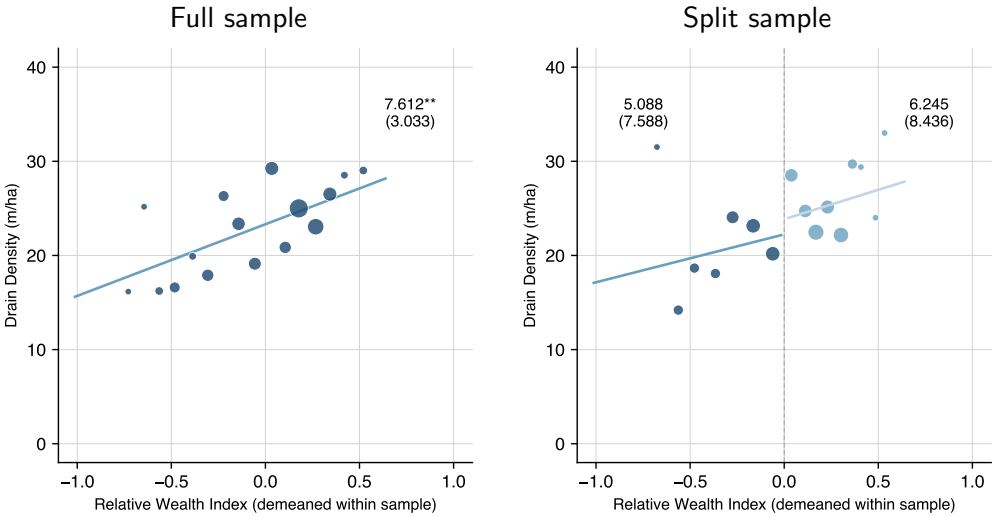
Stormwater drains



Density



Infrastructure and wealth across space



Next steps

- We need more work on adaptation in developing countries
  - Urban flooding, extreme heat, air pollution, sea level rise
- Adaptation policy faces trade-offs